

FORMULAS FOR REFERENCE

SPHERE	Surface area	$= 4\pi r^2$
	Volume	$= \frac{4}{3}\pi r^3$
CYLINDER	Area of curved surface	$= 2\pi rh$
	Volume	$= \pi r^2 h$
CONE	Area of curved surface	$= \pi rl$
	Volume	$= \frac{1}{3}\pi r^2 h$
PRISM	Volume	$= \text{base area} \times \text{height}$
PYRAMID	Volume	$= \frac{1}{3} \times \text{base area} \times \text{height}$

There are 36 questions in Section A and 18 questions in Section B.
The diagrams in this paper are not necessarily drawn to scale.
Choose the best answer for each question.

Section A

Label 1

$$(2x)^5 \cdot x^3 =$$

- A. $6x^6$
- B. $8x^6$
- C. $6x^9$
- D. $8x^9$

Label 2

$$\text{If } 2x - 5y = 7, \text{ then } y =$$

- A. $\frac{5}{2x-7}$
- B. $\frac{5}{2x+7}$
- C. $\frac{2x-7}{5}$
- D. $\frac{2x+7}{5}$

Label 3 $\frac{1}{x+1} - \frac{1}{x-1} =$

- A. $\frac{2}{1-x^2}$
- B. $\frac{2}{x^2-1}$
- C. $\frac{2x}{1-x^2}$
- D. $\frac{2x}{x^2-1}$

Label 4 $pr + qr = ps + qs =$

- A. $(p+q)(r-s)$
- B. $(p+q)(s-r)$
- C. $(p-q)(r-s)$
- D. $(p-q)(s-r)$

Label 5 If $f(x) = \frac{x}{1+x}$, then $f(3)f(\frac{1}{3}) =$

- A. $\frac{3}{16}$
- B. $\frac{1}{2}$
- C. $\frac{3}{4}$
- D. 1

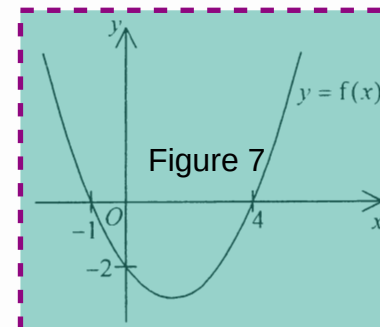
Label 6 Which of the following is an identity / are identities?

- I. $x^2 - 4 = 0$
- II. $x^2 - 4 = (x-2)^2$
- III. $x^2 - 4 = (x+2)(x-2)$

- A. I only
- B. III only
- C. I and II only
- D. I and III only

Label 7 The figure shows the graph of $y = f(x)$. If $f(x)$ is a quadratic function, then $f(x) =$

- A. $\frac{1}{2}(x+1)(x-4)$
- B. $2(x+1)(x-4)$
- C. $\frac{1}{2}(x-1)(x+4)$
- D. $2(x-1)(x+4)$



Label 8 Solve $3x^2 = 21x$

- A. $x = 3$
- B. $x = 7$
- C. $x = 0$ or $x = 7$
- D. $x = 0$ or $x = 3$

Label 9 Find the range of values of k such that the quadratic equation $x^2 + 2x - k = 2$ has two distinct real roots.

- A. $k > -1$
- B. $k \geq -1$
- C. $k \leq -1$
- D. $k \geq 1$

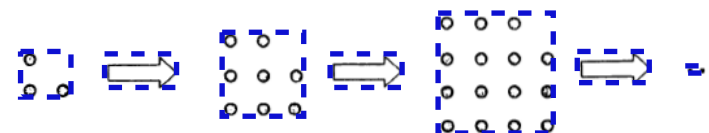
Label 10 The marked price of a car is 50% higher than the cost. If the car is sold at a 20% discount on the marked price, then the percentage profit is

- A. 10%
- B. 20%
- C. 30%
- D. 40%

Label 11 A sum of \$ 14 000 is deposited at 4% per annum for 5 years, compounded yearly. Find the interest correct to the nearest dollar.

- A. \$ 2378
- B. \$ 2800
- C. \$ 1033
- D. \$ 1034

Label 12 In the figure, the 1st pattern consists of 3 dots. For any positive integer n , the $(n+1)$ th pattern is formed by adding $(2n+3)$ dots to the n th pattern. Find the number of dots in the 6th pattern.



- A. 39
- B. 37
- C. 48
- D. 50

Label 13 Let x , y and z be non-zero numbers, if $x:y=1:2$ and $y:z=3:1$, then $(x+y):(y+z)=$

- A. 3:4
- B. 4:3
- C. 8:9
- D. 9:8

Label 14 It is given that x varies directly as y and inversely as z^2 . If y is decreased by 10% and z is increased by 20%, then x is decreased by

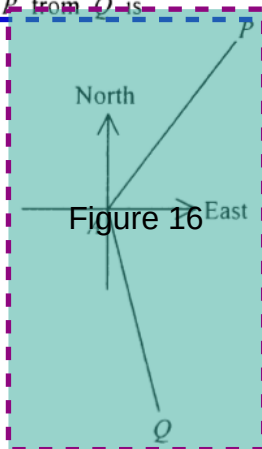
- A. 10%
- B. 23.6%
- C. 25%
- D. 27.5%

Label 15 The scale of a map is $1 : 8\,000$. If the area of a park on the map is 2 cm^2 , then the actual area of the park is

- A. $4\,000\text{ m}^2$
- B. $6\,400\text{ m}^2$
- C. $12\,800\text{ m}^2$
- D. $16\,000\text{ m}^2$

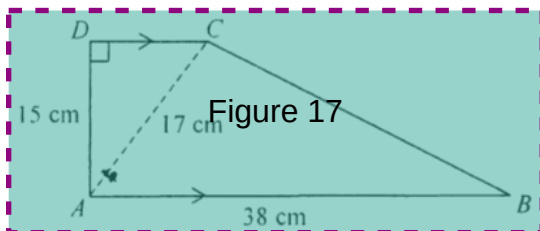
Label 16 In the figure, $PA = QA$. If the bearings of P and Q from A are $N42^\circ\text{E}$ and $S28^\circ\text{E}$ respectively, then the bearing of P from Q is

- A. $N7^\circ\text{E}$
- B. $N27^\circ\text{E}$
- C. $N35^\circ\text{E}$
- D. $N55^\circ\text{E}$



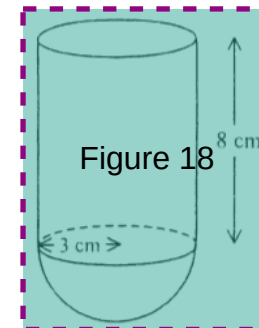
Label 17 In the figure, the area of the trapezium $ABCD$ is

- A. 645 cm^2
- B. 649 cm^2
- C. 690 cm^2
- D. 698 cm^2



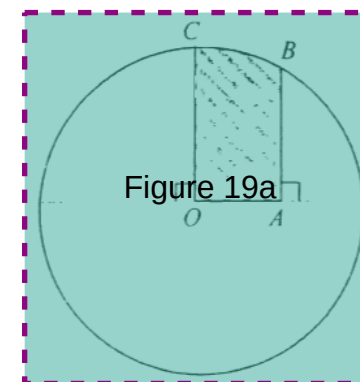
Label 18 In the figure, the solid consists of a hemisphere of radius 3 cm joined to the bottom of a right circular cylinder of height 8 cm and base radius 3 cm . Find the volume of the solid.

- A. $75\pi\text{ cm}^3$
- B. $90\pi\text{ cm}^3$
- C. $93\pi\text{ cm}^3$
- D. $108\pi\text{ cm}^3$

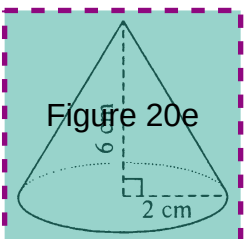
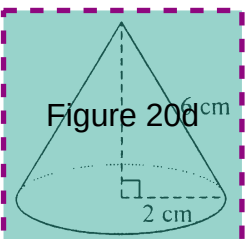
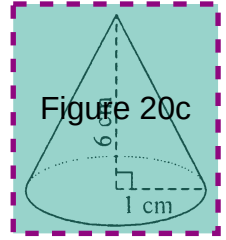
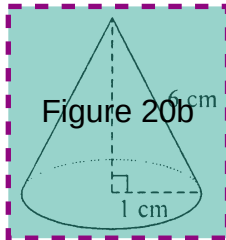
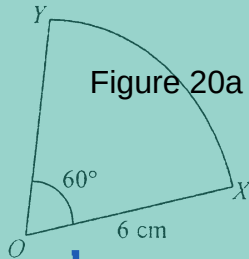


Label 19 In the figure, O is the centre of the circle. B and C are points lying on the circle. If $OC = 2\text{ cm}$ and $OA = 1\text{ cm}$, then the area of the shaded region $OABC$ is

- A. $\frac{\pi}{2}\text{ cm}^2$
- B. $\frac{2\pi}{3}\text{ cm}^2$
- C. $\left(\frac{\sqrt{3}}{2} + \frac{2\pi}{3}\right)\text{ cm}^2$
- D. $\left(\sqrt{3} + \frac{2\pi}{3}\right)\text{ cm}^2$



Label 20 In the figure, sector OXY is a thin metal sheet. By joining OX and OY together, which of the following right circular cones can be folded?



Label 21 $2 \sin(90^\circ - \theta) \sin 60^\circ - \cos 0^\circ \cos \theta =$

- A $\sin \theta$
- B $\sqrt{3} \sin \theta$
- C $\sqrt{3} \cos \theta$
- D $(\sqrt{3} - 1) \cos \theta$

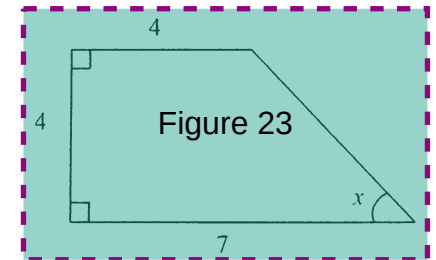
Label 22 If $0^\circ < \theta < 45^\circ$, which of the following must be true?

- I. $\tan \theta < \cos \theta$
- II. $\sin \theta < \tan \theta$
- III. $\sin \theta < \cos \theta$

- A I only
- B II only
- C I and II only
- D I and III only

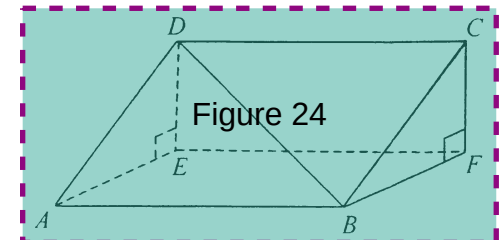
Label 23 In the figure, $\sin x =$

- A $\frac{3}{7}$
- B $\frac{3}{5}$
- C $\frac{4}{5}$
- D $\frac{4}{3}$



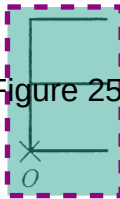
Label 24 The figure shows a right prism $ABCDEF$ with a right-angled triangle as the cross-section. The angle between BD and the plane $CDEF$ is

- A $\angle BDE$
- B $\angle BDF$
- C $\angle DBE$
- D $\angle DBF$



Label 25

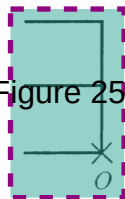
Figure 25a



If the plane figure above is rotated anticlockwise about the point O through 90° , which of the following is its image?

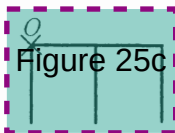
A

Figure 25b



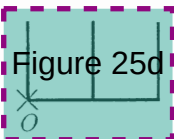
B

Figure 25c



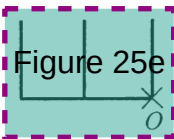
C

Figure 25d



D

Figure 25e



Label 26

In the figure, ABC and AED are straight lines. If $AB = 8$ cm, $BC = 4$ cm and $CD = 9$ cm, then $BE =$

A

$$\frac{32}{9} \text{ cm}$$

B

$$\frac{9}{2} \text{ cm}$$

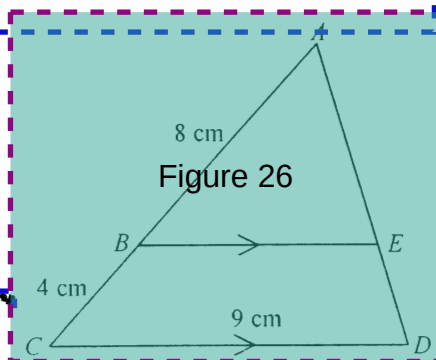
C

$$5 \text{ cm}$$

D

$$6 \text{ cm}$$

Figure 26



Label 27

If the polar coordinates of the points A and B are $(5, 45^\circ)$ and $(12, 135^\circ)$ respectively, then the distance between A and B is

A

$$5$$

B

$$7$$

C

$$13$$

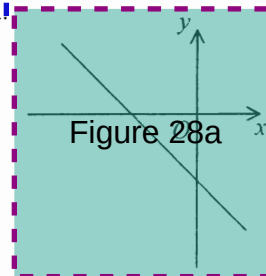
D

$$17$$

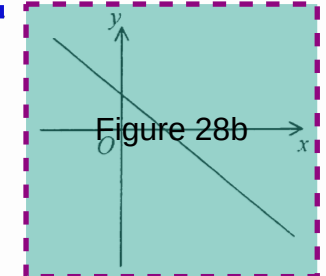
Label 28

If $k < 0$, which of the following may represent the graph of the straight line $x - y = k$?

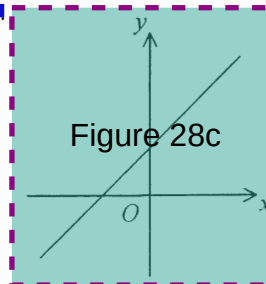
A



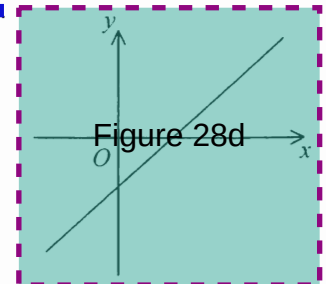
B



C



D



Label 29 The straight line $4x + y - 2 = 0$ is perpendicular to the straight line

- A $4x + y - 9 = 0$
- B $4x - y + 9 = 0$
- C $x + 4y - 9 = 0$
- D $x - 4y + 9 = 0$

Label 30 If the straight line $5x - 3y = 30$ cuts the x -axis and the y -axis at A and B respectively, then the coordinates of the mid-point of AB are

- A $(3, -5)$
- B $(-3, 5)$
- C $(5, -3)$
- D $(-5, 3)$

Label 31 If the points $(0, 0)$, $(2, 0)$ and $(1, b)$ are the vertices of an equilateral triangle, then $b =$

- A 1
- B $\sqrt{3}$
- C 1 or -1
- D $\sqrt{3}$ or $-\sqrt{3}$

Label 32 Which of the following could be the probability of an event?

- A $\frac{\pi}{3}$
- B $\frac{2005}{2006}$
- C -0.2006
- D $\frac{1}{2006}$

Label 33 Two fair dice are thrown. Find the probability that the sum of the two numbers thrown is a prime number.

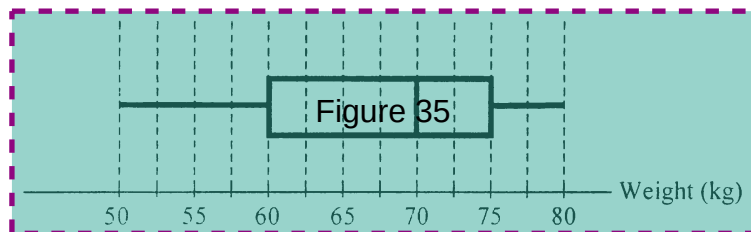
- A $\frac{1}{2}$
- B $\frac{5}{11}$
- C $\frac{5}{12}$
- D $\frac{7}{18}$

Label 34 $\{x-6, x-3, x+4, x+5\}$ and $\{x-8, x-1, x+2, x+9\}$ are two groups of numbers. Which of the following is/are true?

- I. The two groups of numbers have the same mean.
- II. The two groups of numbers have the same median.
- III. The two groups of numbers have the same range.

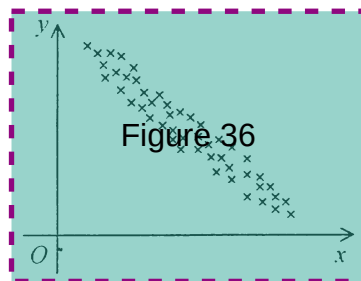
- A I only
- B II only
- C I and III only
- D II and III only

Label 35 The box-and-whisker diagram below shows the distribution of the weights (in kg) of some students. Find the inter-quartile range of their weights.



- A. 5 kg
- B. 10 kg
- C. 15 kg
- D. 30 kg

Label 36 The scatter diagram below shows the relation between x and y . Which of the following may represent the relation between x and y ?

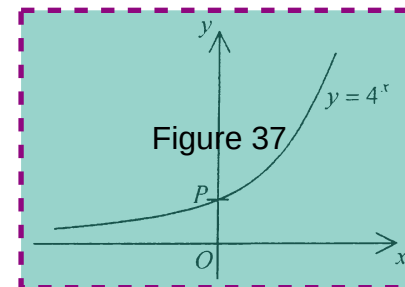


- A. y varies directly as x^2
- B. y decreases when x increases.
- C. x increases when y increases.
- D. x remains unchanged when y increases.

Section B

Label 37 The figure shows the graph of $y = 4^x$. The coordinates of P are

- A. (1, 0)
- B. (0, 1)
- C. (4, 0)
- D. (0, 4)



Label 38 Let a and b be positive numbers. If $\log \frac{a}{10} = 2 \log b$, then $a =$

- A. $10b^4$
- B. $20b$
- C. $b^2 + 10$
- D. $2b + 10$

Label 39 Convert the decimal number $2^{13} + 2^4 + 3$ to a binary number

- A. 10000000000111
- B. 1000000001011
- C. 1000000010011
- D. 100000010011

Label 40 Let k be a non-zero constant. When $x^3 + kx^2 + 2kx + 3k$ is divided by $x + k$, the remainder is k . Find k .

- A. $\frac{1}{2}$
 B. $\frac{1}{3}$
 C. $\frac{1}{4}$
 D. $\frac{1}{5}$

Figure 40

Label 41 In the figure, O is the origin. The equation of AB is $2x + y - 8 = 0$ and the equation of BC is $2x + 3y - 12 = 0$. If (x, y) is a point lying in the shaded region $OABC$ (including the boundary), then the greatest value of $x + 3y + 4$ is

- A. 8
 B. 13
 C. 16
 D. 28

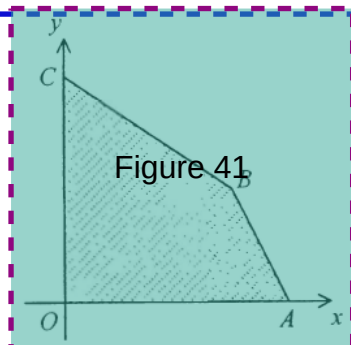


Figure 41

Label 42 The first negative term in the arithmetic sequence 2006, 1998, 1990, ... is

- A. -8
 B. -6
 C. -4
 D. -2

Label 43 Let a , b and c be positive integers. If $b = \sqrt{ac}$, which of the following must be true?

- Label 44 I. $\log a$, $\log b^2$, $\log c$ is an arithmetic sequence.
 II. a^3 , b^3 , c^3 is a geometric sequence.
 III. 4^a , 4^b , 4^c is a geometric sequence.

- A. I and II only
 B. I and III only
 C. II and III only
 D. I, II and III

Label 44 For $0^\circ < x < 360^\circ$, how many roots does the equation $3 \cos^2 x - 4 \cos x + 1 = 0$ have?

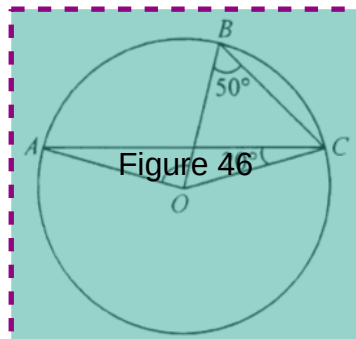
- A. 2
 B. 3
 C. 4
 D. 5

Label 45 If the length of a side of a regular tetrahedron is 3 cm, then the height of the tetrahedron is

- A. 3 cm
 B. $\sqrt{3}$ cm
 C. $\sqrt{6}$ cm
 D. $\frac{3\sqrt{3}}{2}$ cm

Label 46 In the figure, O is the centre of the circle ABC . If $\angle OBC = 50^\circ$ and $\angle ACO = 20^\circ$, then $\angle BQA =$

- A) 50°
- B) 60°
- C) 70°
- D) 80°

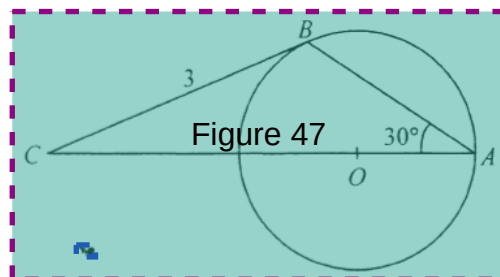


Label 48 Let O be the origin. If the coordinates of the points A and B are $(6, 0)$ and $(0, 6)$, respectively, then the coordinates of the in-centre of $\triangle ABO$ are

- A) $(0, 0)$
- B) $(2, 2)$
- C) $(3, 3)$
- D) $(6 - 3\sqrt{2}, 6 - 3\sqrt{2})$

Label 47 In the figure, O is the centre of the circle. A and B are points lying on the circle. If AOC is a straight line and BC is a tangent to the circle, then the radius of the circle is

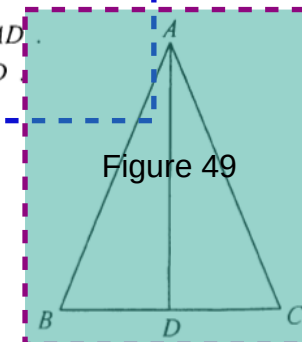
- A) $\frac{3}{2}$
- B) $\sqrt{3}$
- C) $2\sqrt{3}$
- D) $3\sqrt{3}$



Label 49 In the figure, ABC is an acute-angled triangle, $AB = AC$ and D is a point lying on BC such that AD is perpendicular to BC . Which of the following must be true?

- I. The circumcentre of $\triangle ABC$ lies on AD .
- II. The orthocentre of $\triangle ABC$ lies on AD .
- III. The centroid of $\triangle ABC$ lies on AD .

- A) I and II only
- B) I and III only
- C) II and III only
- D) I, II and III



Label 50 Consider the circle $x^2 + y^2 - 4x + 6y - 40 = 0$. Find the slope of the diameter passing through the point $(1, 2)$.

- A. $\frac{1}{3}$
- B. $\frac{1}{5}$
- C. $-\frac{1}{3}$
- D. $-\frac{1}{5}$

Label 51 A circle C cuts the y -axis at A and B . If $AB = 8$ and the coordinates of the centre of C are $(-3, 5)$, then the equation of C is

- A. $x^2 + y^2 + 6x - 10y = 0$
- B. $x^2 + y^2 - 6x + 10y = 0$
- C. $x^2 + y^2 + 6x - 10y + 9 = 0$
- D. $x^2 + y^2 - 6x + 10y + 9 = 0$

Label 52 One letter is chosen randomly from each of the two words 'FORTY' and 'FIFTY'. Find the probability that the two letters chosen are the same.

- A. 0.08
- B. 0.16
- C. 0.32
- D. 0.48

Label 53 There are two questions in a test. The probability that David answers the first question correctly is $\frac{1}{4}$ and the probability that David answers the second question correctly is $\frac{1}{3}$. Given that David answers at least one question correctly in the test, find the probability that he answers the second question correctly.

- A. $\frac{11}{2}$
- B. $\frac{12}{13}$
- C. $\frac{3}{5}$
- D. $\frac{4}{5}$

Label 54 The standard deviation of the five numbers $10a+1$, $10a+3$, $10a+5$, $10a+7$ and $10a+9$ is

- A. 8
- B. $\frac{12}{5}$
- C. $\sqrt{10}$
- D. $2\sqrt{2}$

END OF PAPER